Designing An Online Course

If you are considering teaching online or are looking for ideas to freshen-up your current online course, you have come to the right resource. Designing for the online environment presents unique challenges, but it also opens a world of exciting possibilities for engaging students in their learning. Online education is not an "alternative" to traditional classroom learning. According to a 10 year study conducted by the Online Learning Consortium (http://onlinelearningconsortium.org/survey_report/changing-course-ten-years-tracking-online-education-united-states/), 6.7 million students have taken at least one online course and roughly thirty-two percent of all higher-education students now take at least one online course during their educational career. And these numbers continue to rise. But despite more interest (from students and academic leaders) and enrollment in the online format, student success rates in online classes lags behind their face-to-face counterparts. **So, we cannot afford to ignore the online format, but how do we design effectively for the unique teaching and learning challenges it presents?**

Start with the learning; moving from solutions to possibilities!

It is common when transitioning to the online environment to start with the question, "how can I do this online?" If we approach it from this standpoint, we can get stuck looking for ways to mimic or retrofit face-to-face activities. This can be a frustrating and even **disappointing solution goose chase**. A more productive approach is to start with the question, "what do my students need to learn?". Using the learning as the focal point, you can more easily **navigate the amazing possibilities presented by the online environment**.

The the most effective teaching principles apply regardless of modality and often stand the test of time. Consider the principles identified by 50 years of research by A. Chickering and Z. Gamson in the "Seven Principles for Good Practice in Undergraduate Education." (AAHE Bulletin, March 1987 (http://teaching.uncc.edu/learning-resources/articles-books/best-practice/education-philosophy/seven-principles)):

- encourage faculty-to-student interaction
- encourage student-to-student interaction
- promote active learning
- communicate high expectations
- facilitate time on task
- provide rich, rapid feedback

- respect diverse learning

Although the study was conducted in 1987, these principles identified are amongst the most frequently referenced by online course designers as best practices. Keep these strategies in mind as you examine and approach your course design.

Actualizing Best Practice

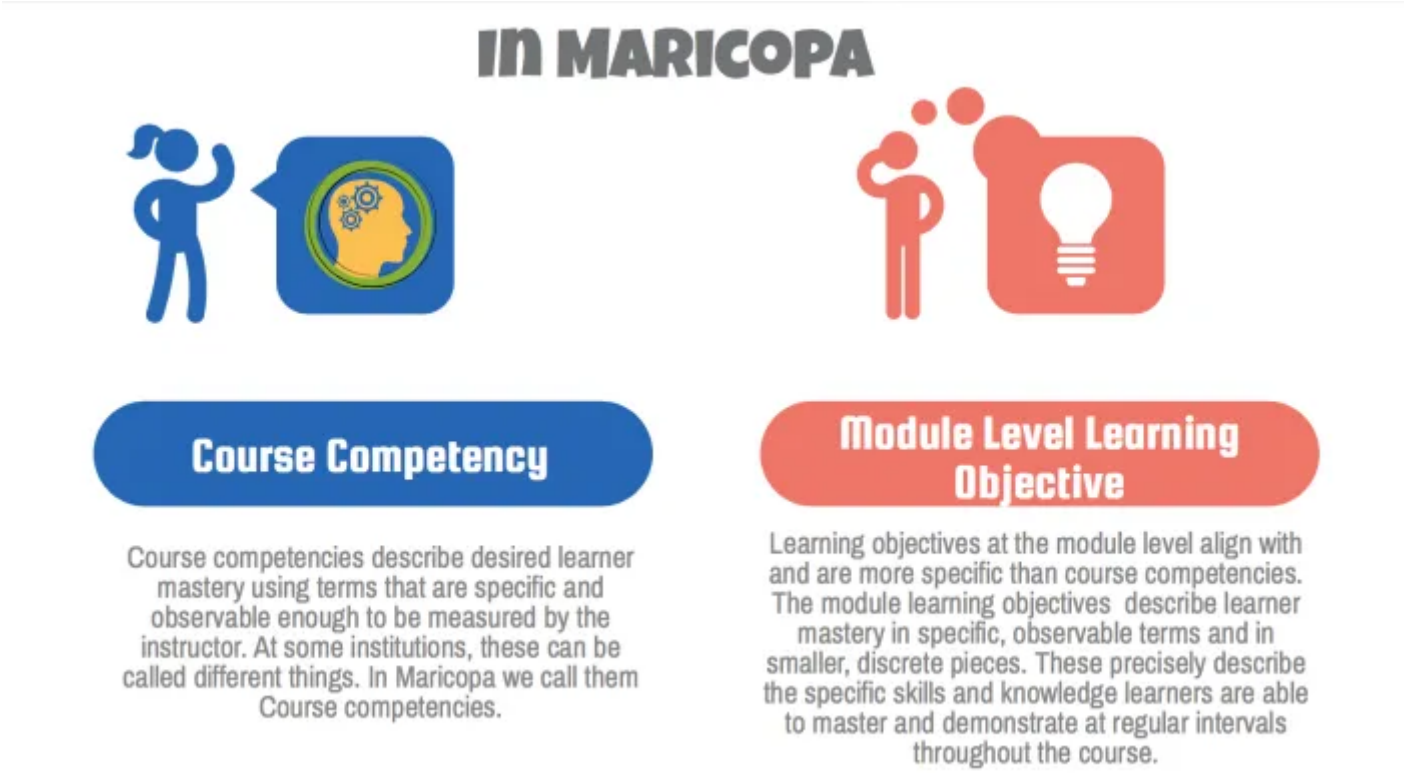
Before you start! Collect all the materials you use to teach your course.

Organization is key to any project. Gather all the resources from previous courses you have taught, content and instructional materials you have researched or picked-up from colleagues, etc.. Put them in a format/file and store in a way you can easily access (computer, online or USB Drive). This includes your syllabi, notes, textbook, lectures, hand-outs, quizzes, exams, assignments/papers/projects, online resources, journal articles, and any other pertinent resources. Ask your department or colleagues for resources specific to your subject area. Sample syllabi, lessons and even course templates may be available to get you started. Taking the time to organize up front will save you lots of time later, so don't skip this critical part of the process.

Establish a timeframe and goals.

Designing a quality course takes time. You need to dedicate constructive and uninterrupted time to plan, design and build. How long it takes is a function of where you are starting (new design or redesign) and your other obligations (work, family, etc.). To ensure your success, establish a realistic timeframe and set goals/benchmarks and deadlines.

In Maricopa there are "course level" competencies that are designed and written by faculty at the district level through the Instructional Council for each discipline. Course competencies are what is required to be covered and taught in every course. For course design and mapping, especially online, a faculty member designs activities, assessments, lectures, etc. to teach those competencies….along the way students will learn incrementally – those are unit or module level learning objectives. These help students understand what they will be learning, how all the activities and assignments help them learn, and then in the end see where they have been.

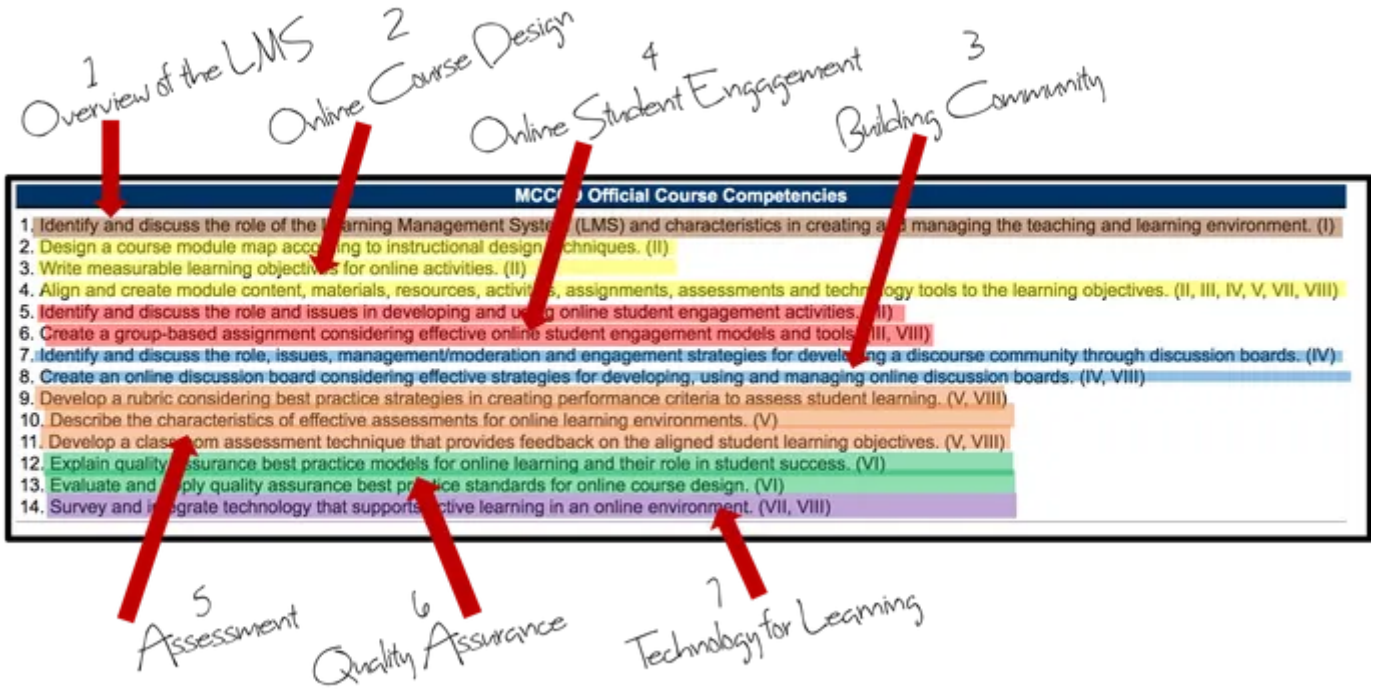


(https://i1.wp.com/ctl.mesacc.edu/wp-content/uploads/2018/06/competency-versus-objective-in-maricopa.png)

Design Practice #1: Course Mapping

Access and review the most current course competencies on the MCCCD Curriculum & Transfer website (https://asa.maricopa.edu/departments/center-for-curriculum-transfer-articulation).

1. Which competencies will be threaded throughout the entire course?
2. Cluster competencies that are relatable. Sequence these clusters into a logical order based on your background knowledge and experiences in the discipline. Module/unit duration is flexible based on the depth of knowledge students might acquire with each topic.
3. What is the theme of each competency cluster? Choose 5-7 units for your course that encompass the competency clusters.



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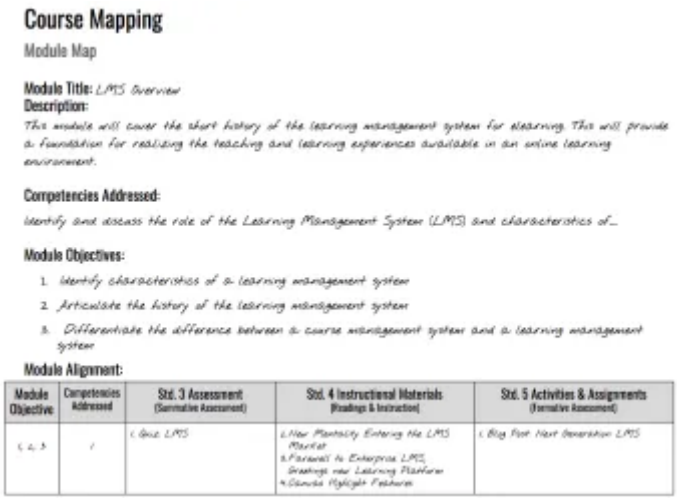
Design Practice #2: Module Map (Key Concepts, Activities, Resources, Vocabulary, and Assessment)

Access and review the most current course outline on the MCCCD Curriculum & Transfer website (https://asa.maricopa.edu/departments/center-for-curriculum-transfer-articulation).

1. Select a module/unit (competency cluster). Which key concepts will be emphasized in the course design?
2. For each key concept, which activities and resources will you integrate for students to make-meaning of course content?
3. Which academic (https://drive.google.com/file/d/0B8TPyPo8yAepWV9wb2cwM09wREk/view) and technical (discipline-specific) terms will you emphasize in each module/unit?

Module Map Planning Example

Map Outline



Course Mapping

Module Map

Module Title: LMS Overview

Description:

This module will cover the short history of the learning management system for elearning. This will provide a foundation for realizing the teaching and learning experiences available in an online learning environment.

Competencies Addressed:

Identify and discuss the role of the Learning Management System (LMS) and characteristics of...

Module Objectives:

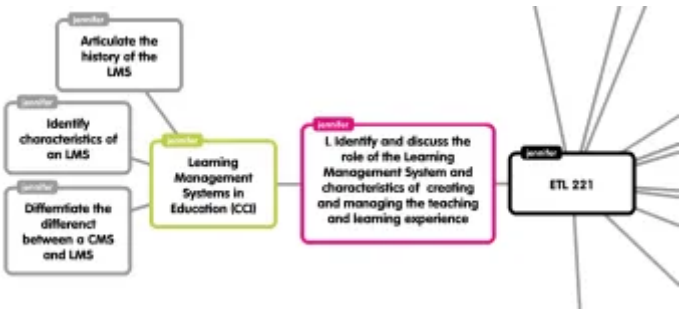
1. Identify characteristics of a learning management system
2. Articulate the history of the learning management system
3. Differentiate the difference between a course management system and a learning management system

Module Alignment:

Module Objective	Competencies Addressed	Std. 3 Assessment (Summative Assessment)	Std. 4 Instructional Materials (Readings & Instruction)	Std. 5 Activities & Assignments (Formative Assessment)
1, 2, 3	1	1 Quiz LMS	1. How Mentality Entering the LMS Market 2. Farewell to Enterprise LMS, Breaking new Learning Platform 4. Canvas Highlight Features	1 Blog Post Next Generation LMS

(http://ctl.mesacc.edu/wp-content/uploads/2018/06/Sample-QM-Module-Map.png)

Concept Map



(https://i1.wp.com/ctl.mesacc.edu/wp-content/uploads/2018/06/Sample-Mind-Map-of-ETL221-LMS-objectives-.png)

Plan for interaction! Make sure your course is rich in opportunities for students to engage with the content, with you and with each other. This means creating diverse activities like discussions, group work, case studies and collaborative problem-solving. Also, be sure to **select resources that are relevant and present a variety of viewpoints and meet different learning styles**. Consider multimedia, periodicals, web resources, etc.

Tip: Don't get too caught up in specifics. This map should be an outline of the unit NOT the individual lessons.

University of Michigan, Center for Research on Teaching & Learning; Active Learning Strategies (http://www.crlt.umich.edu/tstrategies/tsal)

North Carolina Index of Learning Styles Questionnaire (http://www.engr.ncsu.edu/learningstyles/ilsweb.html)

Design Practice #3: Module Level Learning Objectives

Using Course Competencies, pick a module and write learning objectives for one or more module/unit.

Considerations (https://docs.google.com/document/d/1gR8szc07XaLyMYI9mgC_2u8XSLM_-53WZ1KLdT-ape8/edit?usp=sharing)

- Does it begin with an action statement?
- Is it clean and direct?
- Does it express what we can expect students to be able to do?
- Is it clear what we mean?
- Does it reflect the essence of the cluster?

Creating Learning Objectives

It is essential to build measurable and clear objectives that outline what is expected of the learner. These objectives will make it easy to align the rest of your course and will serve to communicate learning expectations to students.

Using Blooms Taxonomy to find Measurable Verbs – Benjamin Bloom and a committee of colleagues identified three domains of learning and objectives can be written for any type of learning (Skills, Knowledge and Attitude). Bloom, and later Anderson & Krathwohl, also outlined categories of thinking. Learning objectives should encourage students to reach higher orders of thinking through careful scaffolding of concepts (structuring learning to build on prior concept knowledge). Using actionable verbs you can create objectives that target learning within these learning types and categories. It is very important that these verbs are **measurable** so that you can assess whether students understood the concept(s). For example:

Non-measurable verbs: understood, appreciate, learn.

Measurable verbs: explain, discuss, compare, etc.

ABCD Method – An easy framework for creating learning objectives is the A.B.C.D. method. This stands for Audience, Behavior, Condition and Degree. Learning objectives that contain each of these elements will clearly outline the learning that is to be achieved after completing each module.

Watch this brief video to learn more. (https://youtu.be/gskWFx39EQg)

Tip: Each module should have approximately 3-5 learning objectives. If you have more, your objectives may be too task oriented or your module theme could be too broad.

Resources:

Use this Blooms Taxonomy Levels Guide (https://drive.google.com/file/d/0B8MKcD1rcHnkRjBESTlPU2RTS2s/view?usp=sharing) to find verbs for your learning objectives. This handy spreadsheet also includes ideas for assignments, activities and assessments that align with these verbs.

Penn State University offers a wonderful interactive tutorial to create objectives according to the ABCD method (http://www.personal.psu.edu/bxb11/Objectives/).

Use the University of Central Florida Objective Builder to create measurable objectives according to the ABCD method (https://online.ucf.edu/teach-online/resources/objective-builder/).

Design Practice #4: Align your module (activities, assignments, materials/technology and assessments) to the objectives.

We all love our course content! As a result, it is tempting to throw everything into your course in an effort to spark that same love in the heart of your students. **But how do students know what is important and essential to their learning? How do we help them focus?**

Achieving Alignment through a Conceptual Framework

Using the learning objectives you can become more selective in what you include in your course. **This selection process is known as alignment.** Alignment occurs when the course component (activity, assignment, material, technology and/or assessment) will help the student meet the learning objectives. To get started, **build a Conceptual Framework for each module**. In this framework outline the learning by identifying the course competencies and learning objectives for the module. Then review the course components (each piece of your module that you identified in the module map process) and see if they fit (align), i.e. contribute to the student achieving the stated learning objective. If a component does not align you need to either change the objective, change the course component or if it is essential to keep this non-aligned component, make sure that it is clearly identified as supplemental.

Tip: Make sure that your framework is as detailed as possible. This framework can be used to provide learners with an overview of each unit that includes what they are to learn (competencies/objectives), with what (activity, assignment, material, technology and/or assessment) and where (in class or online). As an option, you can add a sequence and timeframe to your framework and you have outlined a complete module schedule for your students!

Resources: Use the Module Conceptual Framework Form (https://docs.google.com/document/d/1MBICr8uqjC_JYJk6pG2xHOL7vdopCls8nn6uZFbtrK4/edi usp=sharing) for a course to check your module alignment.

Design Practice #5: Assessment for Learning

You are almost ready to start putting your course online! But before you do, **you need to consider how you will assess student learning**. Assessment is more than just tests, quizzes and final projects. Truly "informative" assessment helps students measure their progress and helps to guide your instruction. How will you embed informal and formal assessments for students to demonstrate understanding of major course concepts?

Summative and Formative

Assessments come in two varieties, summative and formative. **Summative Assessment** evaluates student learning, skill and academic achievement at the end of a defined instructional period (i.e. project, unit, course, semester, etc.). **Formative Assessments** monitor student learning through formal and informal processes to gather evidence to improve learning (i.e. guiding learning from concept to concept, activity to activity and lesson to lesson; identifying clarifications and misconceptions before moving on to the next concept).

C.A.T.s (Classroom Assessment Techniques)

One highly effective type of Formative Assessment is called a C.A.T. (Classroom Assessment Technique). These serve two main purposes, 1) assesses how well your students are learning the content and 2) provides invaluable feedback to guide instruction. C.A.T.s also serve to regularly check that your students are participating and comprehending the content before they get to a Summative Assessment. Regular and purposeful use of C.A.T.s allows the learner to apply and practice what is taught and keeps them engaged in the course more frequently. These opportunities directly contribute to student success and retention.

Tip: Well designed C.A.T.s include a planning, implementing and responding phase.

Rubrics

A rubric is a coherent set of criteria for students' work that includes descriptions of levels of performance quality on the criteria. The main purpose of rubrics is to assess performances (Brookhart, 2013).

Explore four types of rubrics (https://docs.google.com/presentation/d/15FqalLpl2nkZoDtv8J7Uh3RlxpbqJ3B9Q6lHsDMsWz0/ usp=sharing) that you might use to assess assignments in your course. These teacher-created rubrics provide an objective framework to assignments that may lend to subjective review. Select a rubric type that fits your instructional style.

If you are assessing MCC's 4Cs in addition to grading the assignment, import the scoring guidelines into your course and attach the scoring guideline descriptors to your rubric.

Rubric Templates to Import into Your Canvas Course

- Step 1: Download the Canvas Export Package (https://learn.maricopa.edu/courses/1008475/files/46058247/download?wrap=1) for the template.
    - Export Package – *Note: Do not rename or unzip this package file.*
- Step 2: Go to the course you want to import the rubrics into and access the settings from the course navigation.
- Step 3: Select the "Import Content into this Course" option from the right-side menu.
- Step 4: From the Content Type menu select "Canvas Course Export Package".
- Step 5: Choose a file and locate the template export package. *Note: The file extension will end with .imscc.*
- Step 6: Choose the "Select specific content" radial option.
- Step 7: Ignore the date settings and select Import.
- Step 8: When the file has run, choose the Select Content option.
- Step 9: Under the Rubrics area select each rubric type. Leave all other items unchecked.
- Step 10: Click the Select Content button.

Desired Characteristics of Criteria/Descriptors for Rubrics

- **Appropriate:** Represents a competency or learning outcome
- **Definable:** Clear to the instructor and student
- **Observable:** Quality can be seen or heard
- **Distinct from one another:** Each criterion identifies a separate aspect of the learning outcome
- **Complete:** All criteria together describe the whole of the learning outcome
- **Able to support descriptions along a continuum:** Each criterion can be described over a range of performance levels

Ready. Set. GO! Start building your course.

Congratulations the hard work is done! No really, it is! You have mapped your course and your modules, created measurable objectives, aligned all your course components (activities, assignment, materials/technology and assessments) and designed meaningful and varied assessments. **So now**

what? You are ready to put your course online! Let's look at strategies to purposefully approach this process.

Strategy 1: Look at Sample Courses

Examine other online courses to see the features and design elements that you think serve your learning goals. Most faculty like to start by viewing a course in their own discipline, but don't stop there. You'll get a wealth design ideas from courses regardless of the subject area. Good design is not content specific.

MCC Courses

MCC provides previews a sampling of online classes at our eLearning Site (http://www.mesacc.edu/elearning/course-previews). This is a great place to start.

Canvas Courses

Canvas is our Learning Management System (LMS) for MCCCD. View courses from a variety of colleges that use our LMS Canvas in the Canvas Catalog (https://www.canvas.net/). For even more course samples using Canvas explore by feature (http://guides.instructure.com/m/15170).

Open Source Courses

Go beyond Canvas to discover even more amazing course designs.

- **Merlot** (http://www.merlot.org) provides peer reviewed online teaching and learning materials.
- **edEx Courses** (http://www.extension.harvard.edu/open-learning-initiative)are made available by the Harvard Extension School's Open Learning Initiative. Featuring Harvard faculty, the noncredit courses are open to the public. You do not need to register to view the lecture videos.
- **The MIT Open Courseware Initiative** (http://ocw.mit.edu/index.htm) makes MIT course materials that are used in the teaching of almost all undergraduate and graduate subjects available on the web, free of charge, to any user anywhere in the world.

Strategy 2: Identify Quality & Aligned Content Materials

You may want to find additional resources to supplement your own content. The key is make sure that all of your content (regardless of source) aligns with the learning objectives as outlined in Step 3.

Open Educational Resources & the Creative Commons

Open Educational Resources (OER) are freely accessible, open licensed, teaching and learning materials. There are worldwide repositories for the sharing and use of OER. Materials are available in almost any subject area and can include a single image, assignment or activity OR a full textbook and even an entire course.

OER Commons (https://www.oercommons.org/)

Community College Consortium for Open Educational Resources (http://oerconsortium.org/)

Creative Commons resources are less specific and include a variety of resources (educational in purpose or not) that can be used under specific more open licensing arrangements than traditional copyright process. Items include clip art, images, videos, music and more.

Find CC-licensed Works (http://search.creativecommons.org/)

Publisher Content

Publishers often create online courses and course materials that go with your textbook. Talk to your publisher to receive access to the content. Often, you can select the materials and customize it to reach your learning objectives.

Strategy 3: Going beyond Accessibility; Engage in Universal Design

Regardless of where you get your content from, it needs to be accessible. But more than that, your content needs to be Universal. Universal Design or UDL is a set of best practices to help instructors meet the needs of all learners. From designing pages that can be read by mobile devices to screen readers, UDL is about considering the wide diversity of learner needs, not just abilities.

MCC Center for Teaching & Learning, Creating Accessible Course Materials (http://ctl.mesacc.edu/resources/accessibility/)

Strategy 4: Using Canvas

Canvas is the MCCCD adopted Learning Management System (LMS). It is called a learning management system because the **focus is on the facilitation of learning**, not on the storage of content. You will find that Canvas provides wonderful opportunities to enrich the online learning environment, including a built-in multimedia tool, the Edu Apps Center, quizzes, discussions, group and peer review, collaborative documents and so much more!

Canvas is designed to support modules! So, all that hard work you have done mapping out each of your learning units will pay off now. To get started with Canvas take advantage of one of the many opportunities to learn how the system works.

Canvas 101 is an open enrollment course (https://training.instructure.com/courses/179444/) that will walk you through the basics of setting up a new course in Canvas. To enroll, simply click on the "Join this course" button on the Course Home Page. There is no facilitator, but it is a good series of self exercises.

Canvas 102 Demo Videos Pt 1-3 (http://guides.instructure.com/m/4210/l/147167), a comprehensive over 3 hour presentation of all things Canvas for instructors.

The Canvas (http://guides.instructure.com/) Guides (http://guides.instructure.com/) (software developer guides) cover all major features of Canvas by question topic. They are easy to navigate and mostly image-based walk thru demonstrations of how to use a particular feature.

The MCC CTL Getting Started with Canvas in 10 Steps Guide (https://docs.google.com/a/mesacc.edu/document/d/1U_6NDcbPLr8jS0VtPNB3612Rh0T9Wusp=drive_web) is basic primer to get a first time user off the ground.

The MCC Center for Teaching and Learning also offers in-person training for teaching with Canvas. Please visit our calendar (http://ctl.mesacc.edu/events/calendar/) for information on upcoming learning opportunities. Stay up-to-date with all things Canvas by visiting the CTL LMS News (http://ctl.mesacc.edu/resources/canvas/).

Resources for Students – If you are looking for Canvas Guides to assist your students, point them to the Canvas Tutorials and 101 Course (http://ctl.mesacc.edu/resources/canvas/canvas-tutorials/). This course will walk students through all the major features of Canvas making it easy for you to focus on teaching, rather than troubleshooting technology and navigation.

Strategy 5: Getting Started Module

MCC has created a "start here" template module for instructors to import into their courses. Once you bring template in to your Canvas course, you can modify it to meet your specific needs. Instructions on how to import the template are included within the resource.

- Step 1: Download the Canvas Export Package (https://drive.google.com/file/d/10ocCf6QYYH_dpfs37I3Zs1qrViU7ljWc/view?usp=sharing) for the template.
    - Export Package :
      Note: Do not rename or unzip this package file.
- Step 2: Go to the course you want to import the module to and access the settings from the course navigation.

- Step 3: Select the **"Import Content into this Course"** option from the right-side menu.
- Step 4: From the **Content Type menu** select **"Canvas Course Export Package"**.
- Step 5: Choose a file and locate the template export package. *Note: The file extension will end with .imscc.*
- Step 6: Choose the **"Select specific content"** radial option.
- Step 7: Ignore the date settings and select **Import**.
- Step 8: When the file has run, choose the **Select Content** option.
- Step 9: Under the Modules area **select the Getting Started Module. Leave all other items unchecked.**
- Step 10: Click the **Select Content** button.

Final Thoughts

Relax! You won't be creating the perfect online course, at least not the first time you teach it. It takes teaching an online course a few semesters to improve and enhance it. Continue to experiment with new approaches, refining your teaching according to your learning objectives and the feedback of your students. We highly recommend including a Course Survey in your final module so that you can collect valuable data and insights from the student perspective on the design, content and delivery of your course. Give yourself permission to be a student too – to learn over time. This guide will help you create a very good online course to start with. You can improve it from there!

Design + Delivery = Learning

Even a well-designed course can fall short without purposeful delivery throughout the course. Delivery includes pacing, feedback, communication, monitoring and adjusting instruction, etc.. that will actively engage and support the diverse learner needs. Continue your professional development by researching and exploring resources on best practices on instructional delivery.

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